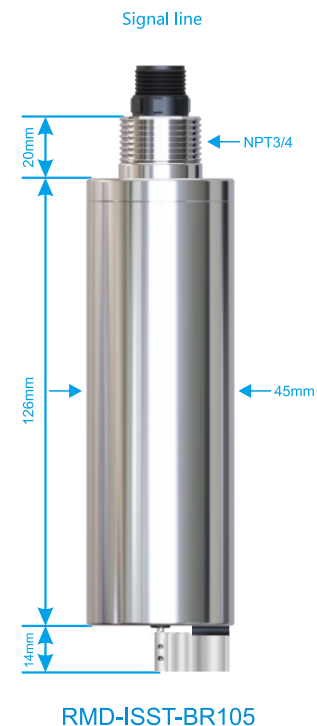


1. Technical data

Principle	Infrared spectral absorption
Measuring range	0...100NTU; 0...200NTU 0...400NTU; 0...1000NTU; 0...4000NTU
Accuracy	±5% FS
Resolution	0.001NTU; 0.01NTU; 0.1NTU
Temperature range	0...60°C
Temperature compensation	Automatic
Lowest detection point	1NTU
Power supply	DC12 ...24V
Output signal	RS485(MODBUS), 4...20mA
Calibration method	Formazin standard solution
Calibration interval	Once every 3-6 months
Shell material	316 Stainless steel
Pressure range	0...3bar
Thread	NPT3/4
Cable length	5m (default) or customized
Self-cleaning brush	YES
Protection grade	Ip68
Installation	Immersion type

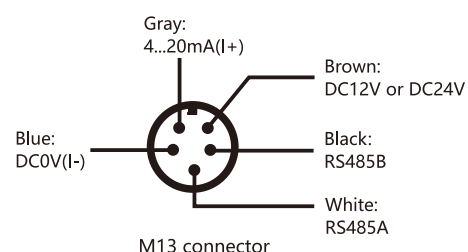
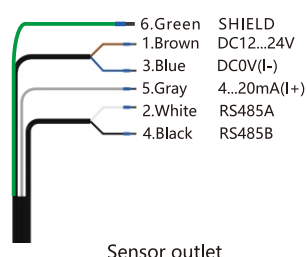


2. Before use

- 2.1 Please read this instruction carefully before use. The probe head cannot be repaired if it is damaged.
- 2.2 In the measurement process, if there is dirt, adhesive or encrust on the probe, the measured value would be inaccurate or fluctuate. The probe should be cleaned and calibrated in time.
- 2.3 The sensor has been calibrated before shipment. To ensure measurement accuracy, the sensor should be re-calibrated before use, if there is a significant error, the sensor should be replaced promptly.
- 2.4 Sensor need to be calibrated regularly, it is recommended to calibrate the sensor every 1 to 2 months.

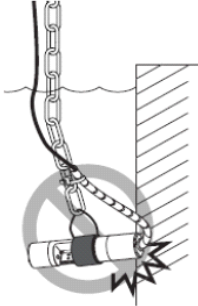
3. Sensor wiring

- 3.1 Please follow the instructions carefully, The wrong wiring will damage the product completely.
- 3.2 Please carefully check all the wiring in the system and confirm that the wiring is complete right before switch on the power.
- 3.3 Note: RS485A line and RS485B line are strictly forbidden to contact with the power supply line, otherwise the communication of the sensor will be permanently damaged.

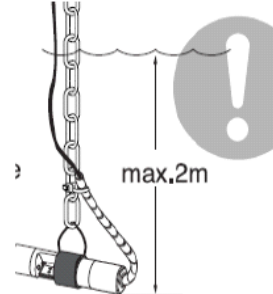


4. Sensor installation

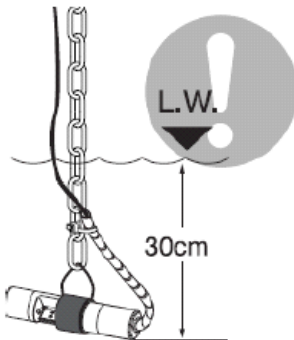
When hanging the sensor, avoid the sensor hitting the wall or other water conservancy facilities caused by water flow, If the current is very strong, fix the sensor



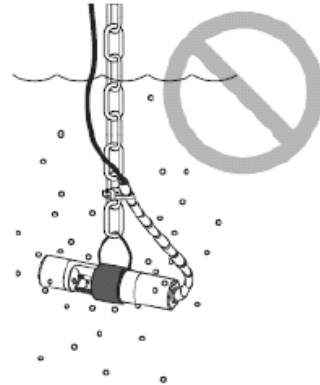
Install the sensor no more than 2 meters from the water surface



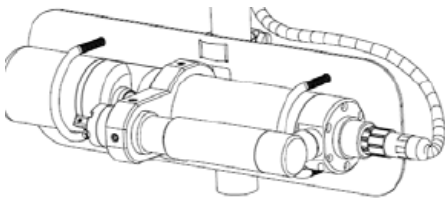
Considering the fluctuation of water level, submerge the sensor below the lowest possible water level of 30CM



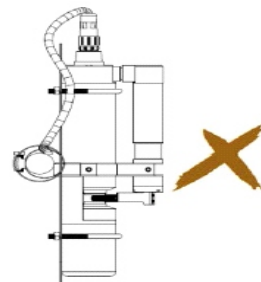
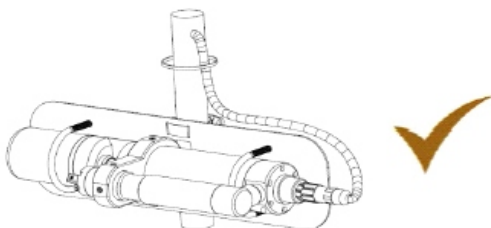
The sensor is placed where there are no bubbles in the water



Install a cable protection sleeve outside the sensor cable, fix the sensor as shown below:



Fix the sensor horizontally:



Note:

Please install the cable protection sleeve properly. Otherwise, it is easy to cause cable damage and water inlet during maintenance. Do not use the sensor cable to lift the sensor.

5. Sensor communication

5.0 Default communication instructions:

- Note: 1. Data starting at 0x represents hexadecimal;
 2. The check code is 16CRC, the low byte is in the front and the high byte is in the back;
 3. Floating point number occupy four bytes;

5.1 Communication description (factory default):

Factory default	
baud rate	9600(default)
data bit	8
stop bit	1
check bit	no
address	1 (default)

5.2 Host computer transmission format:

	Data type	Description	Remarks
Integer	AB (12)	/	Example: 0x 0050 means 80
Floating point number	(CDAB) 3412	Inverted front and back bytes	Example: E1 95 42 7E transposition 42 7E E1 95, floating point number converted to 63.72

5.3 Function code description

5.3.1 This product supports 03,06,16 and other common function codes

5.3.2 The output register uses 16 function codes when writing double word data or writing multiple data in batches

03	Read single or multiple registers
06	Write single register
16	Write multiple registers

5.4 Read floating point number

5.4.1 Host computer transmission format:

	ID address	Function code	Register start address		Qty of registers		CRC16	
			High byte	Low byte	High byte	Low byte	Low byte	High byte
Example 1 Read measured value	0x 01	0x 03	0x 00	0x 01	0x 00	0x 02	0x 95	0x CB

5.4.2 Slave computer response format:

	ID address	Function code	Qty of bytes	Data content				CRC16	
				C	D	A	B	Low byte	High byte
Example Measured value return	0X01	0X03	0X04	0XE1	0X95	0X42	0X7E	0X6C	0XA3

Note:

Data content is measured value: 0X E1 95 42 7E transfer to floating point number, CDAB change order is ABCD, ie 0X 42 7E E1 95 transfer to floating point is 63.72NTU.

5.5 Write floating point number

5.5.1 Host computer transmission format:

	ID address	Function code	Register start address		Qty of registers		Qty of bytes	Data content				CRC16	
			High byte	Low byte	High byte	Low byte		C	D	A	B	Low byte	High byte
Example Write measure upper limit	0X01	0X10	0X00	0X49	0X00	0X02	0X04	0X00	0X00	0X43	0X96	0X86	0XAB

5.5.2 Slave computer response format:

	ID address	Function code	Register start address		Qty of registers		CRC16	
			High byte	Low byte	High byte	Low byte	Low byte	High byte
Example Write upper limit return	0x 01	0x 10	0x 00	0x 49	0x 00	0x 02	0x 90	0x 1E

Note: Convert 300 to hexadecimal, 43 96 00 00 00, transpose to 00 00 43 96, then write to register 0X00 49.

5.6 Modify device address

	ID address	Function code	Register start address		Write register data in hexadecimal integer		CRC16	
			High byte	Low byte	A	B	Low byte	High byte
Example Write device address	0x 00	0x 06	0x 00	0x 00	0x 00	0x 01	0x 49	0x DB
Example Device address return	0x 00	0x 06	0x 00	0x 00	0x 00	0x 01	0x 49	0x DB

Note: Modify device address to 1.

5.7 Two point calibration (e.g. 0 and 100)

5.7.1. Prepare zero point standard solution (distilled water or deionized water) and high point standard solution (e.g. 100NTU)

5.7.2. Set the first point standard value to 0

Send: 01 10 00 24 00 02 04 00 00 00 00 F0 44; Receive: 01 10 00 24 00 02 01 C3

5.7.3. Set the second point standard value to 100

Send: 01 10 00 28 00 02 04 00 00 42 C8 C1 27; Receive: 01 10 00 28 00 02 C1 C0

5.7.4. 1 Place the sensor in the zero point standard solution, send multiple commands to read the AD value, wait for the AD value to stabilize. (about 3 minutes)

Send: 01 03 00 13 00 02 35 CE; Receive: 01 03 04 00 00 3F 80 EA 63

5.7.4. 2 Write the read AD value to the first point standard AD value

Send: 01 10 00 26 00 02 04 00 00 3F 80 61 CD; Receive: 01 10 00 26 00 02 A0 03

5.7.5.1 Place the sensor in 100NTU standard solution, send multiple commands to read the AD value, wait for the AD value to stabilize. (about 3 minutes)

Send: 01 03 00 13 00 02 35 CE; Receive: 01 03 04 00 00 42 DC CB 0A

5.7.5.2 Write the read AD value to the second point standard AD value

Send: 01 10 00 2A 00 02 04 00 00 42 DC 40 F1; Receive: 01 10 00 2A 00 02 60 00

End of calibration

5.8 Address description

(Position) Hexadecimal	Description	Read / Write	Length	Data type	Description
0000	ModBus ID	R/W	1	UINT16	Integer
0001	Measured value	R	2	FLOAT	CDAB
0003	Temperature value	R	2	FLOAT	CDAB
0005	Current output value	R	2	FLOAT	CDAB
0007	Baud rate	R/W	1	UINT16	1=2400; 2=4800 3=9600; 4=19200
0008	Restore factory	W	1	UINT16	Write 1 to restore factory
0009	Cleaning brush	W	1	UINT16	Write 1 execution once
000A	Rotation cycle	R/W	1	UINT16	Unit: minutes
000D	Damping coefficient	R/W	1	UINT16	Integer
0013	Current measured AD value	R	2	FLOAT	CDAB
001A	Firmware version	R	1	UINT16	Integer
0020	Measure offset	R/W	2	FLOAT	CDAB
0022	Temperature offset	R/W	2	FLOAT	CDAB
0024	The first standard value	R/W	2	FLOAT	CDAB
0026	The first standard AD value	R/W	2	FLOAT	CDAB
0028	The second standard value	R/W	2	FLOAT	CDAB
002A	The second standard AD value	R/W	2	FLOAT	CDAB
0049	Measurement upper limit	R/W	2	FLOAT	CDAB
004B	Measurement lower limit	R/W	2	FLOAT	CDAB
004D	Temperature upper limit	R/W	2	FLOAT	CDAB
004F	Temperature upper limit	R/W	2	FLOAT	CDAB
005D	4mA corresponding value	R/W	2	FLOAT	CDAB
005F	20mA corresponding value	R/W	2	FLOAT	CDAB

5.9 Common instruction examples

Send	Receive	Description
00 06 00 00 00 01 49 DB	00 06 00 00 00 01 49 DB	Modify address to 1
01 03 00 01 00 02 95 CB	01 03 04 E1 95 42 7E 6C A3	E1 95 42 7E transposition 42 7E E1 95, floating point number converted to 63.72
01 03 00 03 00 02 34 0B	01 03 04 67 0A 41 82 75 74	67 0A 41 82 transposition 41 82 67 0A, floating point number converted to 16.3
01 06 00 0A 00 3C A9 D9	01 06 00 0A 00 3C A9 D9	Set the rotation cycle to 60 minutes
01 10 00 20 00 02 04 00 00 3F 80 E1 E7	01 10 00 20 00 02 40 02	Measurement value offset +1
01 10 00 22 00 02 04 00 00 3F 80 60 3E	01 10 00 22 00 02 E1 C2	Temperature value offset +1
01 03 00 13 00 02 35 CE	01 03 04 00 00 44 7A 48 D0	Read current measured AD value (read during calibration)
01 10 00 24 00 02 04 00 00 00 00 F0 44	01 10 00 24 00 02 01 C3	The first point standard value, set to 0
01 10 00 26 00 02 04 00 00 00 00 71 9D	01 10 00 26 00 02 A0 03	The first point standard AD value, set to 0
01 10 00 28 00 02 04 00 00 42 C8 C1 27	01 10 00 28 00 02 C1 C0	The second point standard value, set to 100
01 10 00 2A 00 02 04 00 00 44 7A C3 2B	01 10 00 2A 00 02 60 00	The second point standard AD value, set to 1000
01 10 00 49 00 02 04 00 00 43 96 86 AB	01 10 00 2A 00 02 90 1E	Set the measurement upper limit to 300
01 10 00 4B 00 02 04 00 00 00 00 B6 2C	01 10 00 4B 00 02 31 DE	Set the lower measurement limit to 0
01 10 00 4D 00 02 04 00 00 42 70 07 42	01 10 00 4D 00 02 D1 DF	Set the upper temperature limit to 60
01 10 00 4F 00 02 04 00 00 00 00 B7 DF	01 10 00 4D 00 02 70 1F	Set the lower temperature limit to 0
01 10 00 5D 00 02 04 00 00 00 00 37 0A	01 10 00 5D 00 02 D0 1A	Set 4mA corresponding value to 0
01 10 00 5F 00 02 04 00 00 43 96 07 8D	01 10 00 5F 00 02 71 DA	Set 20mA corresponding value to 300

6. Maintenance and storage

6.1 The organic dirt on the sensor can be cleaned with warm water containing detergent or alcohol. After cleaning the sensor, it should be dried by soft tissue.

6.2 When the sensor is stored, it shall be dried and stored .

6.3 Cable connector must be kept clean and free from moisture or water.

6.4 Maintenance rate:

Maintenance task	Recommended maintenance rate
Clean sensor	Clean every 30 days
Check if the sensor is damaged	Check every 30 days

7. Troubleshooting

7.1 The failure rate of the sensor is low, when the measurement is inaccurate, mainly because the sensor has changed, so it is necessary to check whether the sensor is in good condition.

7.2 If the value of the sensor is too large, too small or no change, please check whether the sensor is in good connection with the instrument.

7.3 Modbus troubleshooting:

Problem	Possible reason	Solution
Modbus no response	The baud rate, or stop bit does not match the Modbus master settings	Verify that the settings match the Modbus master device settings, and verify that the Modbus master device parity check is set to None
	Rs232 or RS485 cable is faulty	Replace/repair cables
	No network offsets and terminations, or network offsets and terminations are not suitable.	Check the termination or offset Settings for all network devices. Only the endpoints of the network should be turned on and terminated, and there should be only a point on the network to provide an offset.
	The slave address is incorrect, or the slave address is the same as the address of another bus device	Verify that all addresses are unique and are between 1 and 247.
Modbus abnormal response	Register not supported	Verify that the register is supported
	Incorrect data type	Verify that the requested register data type matches the Modbus master device request; for example, you cannot access a floating point data using 2-byte integer data. When a floating point data (2 registers / 4 bytes) is requested, two registers must be requested at the same time.

8. Warranty

The sensor has a one year warranty period. As long as the damage is caused by improper use of non-human within the warranty period, please prepaid freight, pack the sensor and ship it back, we will repair it for you free of charge. We will analyze the reasons for the damage of the sensor, if the damage exceeds the warranty conditions, we need to charge the repair fee.